

习题 15.1

张志聪

2025 年 11 月 25 日

1

• (1)

– (i)

$f(x)$ 在 $[-\pi, \pi]$ 上是按段光滑的, 可以展开为傅里叶级数

$$\begin{aligned}a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx \\&= \frac{1}{\pi} \int_{-\pi}^{\pi} x dx \\&= \frac{1}{\pi} \int_{-\pi}^{\pi} x dx \\&= 0\end{aligned}$$

$$\begin{aligned}a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \\&= \frac{1}{\pi} \int_{-\pi}^{\pi} x \cos nx dx \\&= 0\end{aligned}$$

注: $x \cos nx$ 是奇函数。

$$\begin{aligned}
 b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx \\
 &= \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin nx dx \\
 &= \frac{1}{\pi} \left(-\frac{1}{n}\right) \int_{-\pi}^{\pi} x d(\cos nx) \\
 &= -\frac{1}{n\pi} \left(x \cos nx \Big|_{-\pi}^{\pi} - \int_{-\pi}^{\pi} \cos nx dx \right) \\
 &= -\frac{1}{n\pi} [\pi(-1)^n - (-\pi)(-1)^n - 0] \\
 &= (-1)^{n+1} \frac{2}{n}
 \end{aligned}$$

于是

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2}{n} \sin nx$$

由于 $f(x)$ 在 $(-\pi, \pi)$ 上连续, 可知

$$f(x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2}{n} \sin nx, \quad x \in (-\pi + 2k\pi, \pi + 2k\pi)$$

由收敛定理可知, $x = \pm\pi$ 时

$$\begin{aligned}
 \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2}{n} \sin nx &= \frac{f(\pi+0) + f(\pi-0)}{2} \\
 &= \frac{\pi + (-\pi)}{2} \\
 &= 0
 \end{aligned}$$

– (ii) 略

• (2)

– (1)

$f(x)$ 在 $[-\pi, \pi]$ 上是按段光滑的, 可以展开为傅里叶级数, 有

$$\begin{aligned}
 a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \\
 &= \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos nx dx \\
 &= \frac{1}{n\pi} \int_{-\pi}^{\pi} x^2 d(\sin nx) \\
 &= \frac{1}{n\pi} \left(x^2 \sin nx \Big|_{-\pi}^{\pi} - \int_{-\pi}^{\pi} 2x \sin nx dx \right) \\
 &= \frac{1}{n\pi} \left(0 - \int_{-\pi}^{\pi} 2x \sin nx dx \right) \\
 &= -\frac{2}{n\pi} \int_{-\pi}^{\pi} x \sin nx dx \\
 &= -\frac{2}{n\pi} \left(-\frac{1}{n} \right) \int_{-\pi}^{\pi} x d(\cos nx) \\
 &= \frac{2}{n^2\pi} \left(x \cos nx \Big|_{-\pi}^{\pi} - \int_{-\pi}^{\pi} \cos nx dx \right) \\
 &= \frac{2}{n^2\pi} [(-1)^n 2\pi - 0] \\
 &= (-1)^n \frac{4}{n^2}
 \end{aligned}$$

$$\begin{aligned}
 a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx \\
 &= \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 dx \\
 &= \frac{1}{\pi} \left(\frac{1}{3} x^3 \Big|_{-\pi}^{\pi} \right) \\
 &= \frac{1}{\pi} \frac{2}{3} (\pi)^3 \\
 &= \frac{2}{3} (\pi)^2
 \end{aligned}$$

$$\begin{aligned}
 b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx \\
 &= \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \sin nx dx \\
 &= 0
 \end{aligned}$$

故

$$\begin{aligned} f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \\ &= \frac{(\pi)^2}{3} \sum_{n=1}^{\infty} (-1)^n \frac{4}{n^2} \cos nx, \quad x \in (-\pi, \pi) \end{aligned}$$

注: $x = \pm\pi$ 时,

$$\begin{aligned} \frac{f(\pi+0) + f(\pi-0)}{2} &= \pi^2 = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} (-1)^n \frac{4}{n^2} \cos n\pi \\ &= \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} \end{aligned}$$

经过变形, 我们有

$$\frac{\pi^2}{6} = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

– (2)

$f(x)$ 在 $[0, 2\pi]$ 上是按段光滑的, 可以展开为傅里叶级数, 有

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_0^{2\pi} f(x) dx \\ a_n &= \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx \\ b_n &= \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx \end{aligned}$$

计算略

• (3)

$$f'(x) = \begin{cases} a, & -\pi < x < 0 \\ b, & 0 < x < \pi \end{cases}$$

又因为

$$\begin{aligned} \lim_{x \rightarrow 0^-} f'(x) &= a \\ \lim_{x \rightarrow 0^+} f'(x) &= b \end{aligned}$$

所以 $f(x)$ 在 $[-\pi, \pi]$ 上是分段光滑的, 可以展开为傅里叶级数

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx \\ &= \frac{1}{\pi} \left(\int_{-\pi}^0 ax dx + \int_0^{\pi} bx dx \right) \\ &= \frac{b-a}{2} \pi \end{aligned}$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \\ &= \frac{1}{\pi} \left(\int_{-\pi}^0 ax \cos nx dx + \int_0^{\pi} bx \cos nx dx \right) \\ &= \frac{a-b}{n^2 \pi} [1 - (-1)^n] \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx \\ &= \frac{1}{\pi} \left(\int_{-\pi}^0 ax \sin nx dx + \int_0^{\pi} bx \sin nx dx \right) \\ &= (-1)^n \frac{a+b}{n} \end{aligned}$$

故

$$\begin{aligned} &\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \\ &= \frac{b-a}{4} \pi + \sum_{n=1}^{\infty} \frac{a-b}{n^2 \pi} [1 - (-1)^n] \cos nx + (-1)^n \frac{a+b}{n} \sin nx \end{aligned}$$